

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Case Study (Medium)

Assessment Tool Weightage: 30%

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QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5

7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5
8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

I _K.SANDHYA 192424061_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

QUESTION : 1. Online Shopping - Normalize to 3NF

Relation:

Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)

Functional Dependencies

- $\text{OrderID} \rightarrow \text{CustID}$
- $\text{CustID} \rightarrow \text{CustName}$
- $\text{ProdID} \rightarrow \text{ProdName, Price}$
- $(\text{OrderID, ProdID}) \rightarrow \text{Qty}$

Assume one order can contain multiple products.

Candidate Key

(OrderID, ProdID)

1NF

All attributes contain atomic values, so the relation is in **1NF**.

2NF

There are partial dependencies:

- $\text{OrderID} \rightarrow \text{CustID}$
- $\text{ProdID} \rightarrow \text{ProdName, Price}$

Therefore, decompose:

- ORDER(OrderID, CustID)
- CUSTOMER(CustID, CustName)
- PRODUCT(ProdID, ProdName, Price)
- ORDER_ITEM(OrderID, ProdID, Qty)

3NF

The transitive dependency is:

$\text{OrderID} \rightarrow \text{CustID} \rightarrow \text{CustName}$

This has already been removed by creating CUSTOMER.

Final 3NF Design

- CUSTOMER(CustID, CustName)
- ORDER(OrderID, CustID)
- PRODUCT(ProdID, ProdName, Price)
- ORDER_ITEM(OrderID, ProdID, Qty)

All non-key attributes depend on the key, the whole key, and nothing but the key.

2. Hospital — Normalize to BCNF

Relation:

Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)

Functional Dependencies

Assume:

- $\text{PID} \rightarrow \text{PName}, \text{DocID}, \text{WardNo}$
- $\text{DocID} \rightarrow \text{DocName}, \text{Specialization}$
- $\text{WardNo} \rightarrow \text{WardName}$

Candidate Key

PID

Anomalies

1. Update anomaly:

If a doctor's specialization changes, it must be updated for every patient of that doctor.

2. Insertion anomaly:

A doctor cannot be stored unless a patient is assigned to that doctor.

3. Deletion anomaly:

Deleting the last patient of a doctor may remove information about the doctor.

BCNF Decomposition

Doctor :

DOCTOR(DocID, DocName, Specialization)

FD:

$\text{DocID} \rightarrow \text{DocName}, \text{Specialization}$

DocID is the key, so BCNF.

Ward :

WARD(WardNo, WardName)

FD:

$\text{WardNo} \rightarrow \text{WardName}$

WardNo is the key, so BCNF.

Patient :

PATIENT(PID, PName, DocID, WardNo)

FD:

$PID \rightarrow PName, DocID, WardNo$

PID is the key, so BCNF.

Final BCNF Design

- PATIENT(PID, PName, DocID, WardNo)
- DOCTOR(DocID, DocName, Specialization)
- WARD(WardNo, WardName)

QUESTION : 3. University Enrollment - Apply 2NF

Relation:

Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade)

Functional Dependencies

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- $(SID, CourseID) \rightarrow Grade$

Candidate Key

- (SID, CourseID)
- 1NF
- All values are atomic, so the relation is in **1NF**.

Partial Dependencies

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- These depend only on parts of the composite key.
- Therefore, the relation is **not in 2NF**.

- 2NF Decomposition

STUDENT

- STUDENT(SID, SName)
- Primary Key: SID

COURSE

COURSE(CourseID, CourseName, Faculty)

Primary Key: CourseID

ENROLLMENT

- ENROLLMENT(SID, CourseID, Grade)
- Primary Key: (SID, CourseID)

Final Answer

- STUDENT(SID, SName)
- COURSE(CourseID, CourseName, Faculty)
- ENROLLMENT(SID, CourseID, Grade)

This removes all partial dependencies and achieves **2NF**.

QUESTION : 4. Airline Booking — Normalize to BCNF

Relation:

Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)

Functional Dependencies

Assume:

- $\text{PassID} \rightarrow \text{PassName}$
- $\text{FlightNo} \rightarrow \text{DeptCity}, \text{ArrCity}$
- $(\text{FlightNo}, \text{Date}, \text{Seat}) \rightarrow \text{PassID}$
- $(\text{FlightNo}, \text{Date}, \text{PassID}) \rightarrow \text{Seat}$

Candidate Keys

Possible candidate keys:

- $(\text{FlightNo}, \text{Date}, \text{Seat})$
- $(\text{FlightNo}, \text{Date}, \text{PassID})$

BCNF Decomposition

PASSENGER

PASSENGER(PassID, PassName)

$\text{PassID} \rightarrow \text{PassName}$

BCNF.

FLIGHT

FLIGHT(FlightNo, DeptCity, ArrCity)

$\text{FlightNo} \rightarrow \text{DeptCity}, \text{ArrCity}$

BCNF.

BOOKING

BOOKING(FlightNo, Date, PassID, Seat)

Candidate keys:

- $(\text{FlightNo}, \text{Date}, \text{PassID})$
- $(\text{FlightNo}, \text{Date}, \text{Seat})$

BCNF under the assumed dependencies.

Final BCNF Design

- PASSENGER(PassID, PassName)
- FLIGHT(FlightNo, DeptCity, ArrCity)
- BOOKING(FlightNo, Date, PassID, Seat)

QUESTION : 5. HR System - Normalize to 3NF

A typical multinational HR relation can be represented as:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName, CountryID, CountryName)

Functional Dependencies

- EmpID $\rightarrow$ EmpName, DeptID, ManagerID, CountryID
- DeptID $\rightarrow$ DeptName, ManagerID
- ManagerID $\rightarrow$ ManagerName
- CountryID $\rightarrow$ CountryName

Candidate Key

EmpID

Transitive Dependencies

We have:

EmpID $\rightarrow$ DeptID $\rightarrow$ DeptName

and

EmpID $\rightarrow$ ManagerID $\rightarrow$ ManagerName

and

$\text{EmpID} \rightarrow \text{CountryID} \rightarrow \text{CountryName}$

These are **transitive dependencies**.

3NF Decomposition

EMPLOYEE

EMPLOYEE(EmpID, EmpName, DeptID, ManagerID, CountryID)

DEPARTMENT

DEPARTMENT(DeptID, DeptName, ManagerID)

MANAGER

MANAGER(ManagerID, ManagerName)

COUNTRY

COUNTRY(CountryID, CountryName)

Result

The transitive dependencies are removed and the schema satisfies **3NF**.

QUESTION : 6. Library - Decompose to 4NF

Relation:

Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)

Functional Dependencies

- $\text{MemberID} \rightarrow \text{MemberName}$
- $\text{BookID} \rightarrow \text{Title, Author, Genre}$
- $(\text{MemberID, BookID}) \rightarrow \text{BorrowDate}$

There can also be independent multivalued facts associated with a member, such as books borrowed and other independent activities.

MVD

Suppose a member can borrow multiple books independently:

MemberID $\twoheadrightarrow$ BookID

If there are independent multi-valued attributes, storing them together causes redundancy.

4NF Decomposition

MEMBER

MEMBER(MemberID, MemberName)

BOOK

BOOK(BookID, Title, Author, Genre)

BORROW

BORROW(MemberID, BookID, BorrowDate)

Why 4NF?

4NF requires that every non-trivial MVD has a superkey as its determinant.

The independent multi-valued facts are separated, eliminating unnecessary combinations and redundancy.

QUESTION : 7. Telecom Billing — Keys and Functional Dependencies

A suitable telecom billing relation can be represented as:

Billing(BillID, CustomerID, CustomerName, PhoneNo, PlanID, PlanName, CallID, CallDate, Duration, Rate)

Functional Dependencies

- $\text{BillID} \rightarrow \text{CustomerID}, \text{PlanID}$
- $\text{CustomerID} \rightarrow \text{CustomerName}$
- $\text{PhoneNo} \rightarrow \text{CustomerID}$
- $\text{PlanID} \rightarrow \text{PlanName}, \text{Rate}$
- $\text{CallID} \rightarrow \text{PhoneNo}, \text{CallDate}, \text{Duration}$
- $(\text{PhoneNo}, \text{CallID}) \rightarrow \text{Duration}$

Candidate Keys

Depending on the business rules:

- BillID
- CallID if each call is uniquely identified
- (PhoneNo, CallID) if CallID is unique only within a phone number

Primary Key

If BillID uniquely identifies a bill:

Primary Key = BillID

Foreign Keys

- CustomerID references CUSTOMER
- PlanID references PLAN
- PhoneNo references PHONE/CUSTOMER
- CallID identifies a call record

Important Dependencies

$\text{BillID} \rightarrow \text{CustomerID}$

$\text{CustomerID} \rightarrow \text{CustomerName}$

Therefore:

$\text{BillID} \rightarrow \text{CustomerID} \rightarrow \text{CustomerName}$

This is a **transitive dependency**.

Similarly:

$\text{BillID} \rightarrow \text{PlanID} \rightarrow \text{PlanName, Rate}$

These dependencies should be considered during normalization.

QUESTION : 8. Retail Inventory — Complete Normalization

Relation:

Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)

Functional Dependencies

- $\text{PID} \rightarrow \text{PName, SupplierID, Category, Price, Warehouse}$
- $\text{SupplierID} \rightarrow \text{SupplierName}$

Candidate Key

PID

1NF

All attributes are atomic $\rightarrow$ 1NF.

2NF

The key is a single attribute PID, so partial dependency cannot exist.

Therefore $\rightarrow$ 2NF.

3NF

We have:

$PID \rightarrow SupplierID$

and

$SupplierID \rightarrow SupplierName$

Therefore:

$PID \rightarrow SupplierID \rightarrow SupplierName$

This is a transitive dependency.

Decomposition

PRODUCT

PRODUCT(PID, PName, SupplierID, Category, Price, Warehouse)

SUPPLIER

SUPPLIER(SupplierID, SupplierName)

Final Design

PRODUCT(PID, PName, SupplierID, Category, Price, Warehouse)

SUPPLIER(SupplierID, SupplierName)

This removes the transitive dependency and produces a normalized design.

QUESTION : 9. School ERP -Normalize up to BCNF

A poorly designed School ERP relation could be:

School(StudentID, StudentName, ClassID, ClassName, TeacherID, TeacherName, SubjectID, SubjectName, Grade)

Functional Dependencies

- $\text{StudentID} \rightarrow \text{StudentName}$
- $\text{ClassID} \rightarrow \text{ClassName}, \text{TeacherID}$
- $\text{TeacherID} \rightarrow \text{TeacherName}$
- $\text{SubjectID} \rightarrow \text{SubjectName}$
- $(\text{StudentID}, \text{SubjectID}) \rightarrow \text{Grade}$

Candidate Key

(StudentID, SubjectID)

Problems

There are partial dependencies:

$\text{StudentID} \rightarrow \text{StudentName}$

$\text{SubjectID} \rightarrow \text{SubjectName}$

There are also transitive dependencies:

$\text{ClassID} \rightarrow \text{TeacherID} \rightarrow \text{TeacherName}$

BCNF Decomposition

STUDENT

STUDENT(StudentID, StudentName, ClassID)

CLASS

CLASS(ClassID, ClassName, TeacherID)

TEACHER

TEACHER(TeacherID, TeacherName)

SUBJECT

SUBJECT(SubjectID, SubjectName)

RESULT

RESULT(StudentID, SubjectID, Grade)

BCNF Justification

In each relation, every determinant is a candidate key:

- $\text{StudentID} \rightarrow \text{StudentName}, \text{ClassID}$
- $\text{ClassID} \rightarrow \text{ClassName}, \text{TeacherID}$
- $\text{TeacherID} \rightarrow \text{TeacherName}$
- $\text{SubjectID} \rightarrow \text{SubjectName}$
- $(\text{StudentID}, \text{SubjectID}) \rightarrow \text{Grade}$

Therefore, the decomposed relations satisfy **BCNF**.

10. Movie Streaming — 5NF

Consider:

Streaming(MovieID, ActorID, LanguageID)

Suppose the system records independent relationships:

- Which actors appear in a movie
- Which languages are available for a movie
- Which actor-language combinations are supported

Join Dependency

The relation may have a join dependency:

$$R = R1 \bowtie R2 \bowtie R3$$

where:

- $R1(\text{MovieID}, \text{ActorID})$
- $R2(\text{MovieID}, \text{LanguageID})$
- $R3(\text{ActorID}, \text{LanguageID})$

If these relationships are independent, storing them together can create unnecessary tuples.

5NF Decomposition

MOVIE_ACTOR

MOVIE_ACTOR(MovieID, ActorID)

MOVIE_LANGUAGE

MOVIE_LANGUAGE(MovieID, LanguageID)

ACTOR_LANGUAGE

ACTOR_LANGUAGE(ActorID, LanguageID)

Lossless Join Verification

The original relation can be reconstructed as:

$$\text{MOVIE_ACTOR} \bowtie \text{MOVIE_LANGUAGE} \bowtie \text{ACTOR_LANGUAGE}$$

If the decomposition generates exactly the original valid tuples and does not introduce spurious tuples, the decomposition is **lossless**.

5NF Condition

A relation is in **5NF** when every non-trivial join dependency is implied by candidate keys.

Thus, independent many-to-many relationships are separated into individual relations, eliminating join dependency redundancy.

Quick Revision Table

Q	Main Topic	Key Result
1	Online Shopping	3NF
2	Hospital	BCNF
3	University	2NF
4	Airline	BCNF
5	HR	3NF
6	Library	4NF
7	Telecom	Keys + FDs
8	Retail	Complete normalization
9	School ERP	BCNF
10	Movie Streaming	5NF

Normalization order to remember:

1NF → 2NF → 3NF → BCNF → 4NF → 5NF

- **1NF:** Atomic values
- **2NF:** Remove partial dependency
- **3NF:** Remove transitive dependency
- **BCNF:** Every determinant must be a candidate key
- **4NF:** Remove unwanted multivalued dependencies
- **5NF:** Remove unwanted join dependencies

